

Multivariate statistics

Lecture 05:

MANOVA, multivariate analysis of variance

Literature and Problems: See course schedule

Agenda (preliminary):

MANOVA 1, One-factor multivariate analysis of variance

- model with MVN populations $l = 1, \dots, g$
- assumptions on covariance matrices and mean vectors
- hypothesis H_0
- estimation of group mean and overall mean
- decomposition of total “variation” (total “sum-of-squares”), $SS_T = SS_W + SS_B$
- MANOVA 1 table: SS, degrees of freedom, distribution
- estimators S_B and S_W for Σ
- **SLIDE: MANOVA 1 test-statistics, definitions, exact and large-sample approximate distributions**
- test of hypothesis
- accept of H_0 : new model and estimation of remaining parameters
- rejection of H_0 : new model and estimation of remaining parameters
- pair-wise Bonferroni CI's for difference in means (total of $\binom{g}{2} p$ 1D CI's)

Model check

- check of residual vectors for being MVN using scatter matrix plot and Mahalanobis distance plot
- Bartlett test for equal group covariance matrices

MANOVA 2, Two-factor multivariate analysis of variance with interactions

- model with MVN populations arranged in “matrix” and classified using factor 1 column index $l = 1, \dots, g$ and factor 2 row index $k = 1, \dots, b$
 - assumptions on covariance matrices and mean vectors
 - illustration of “interaction”
 - estimation of overall mean, column means, row means, and cell means
 - decomposition: $SS_T = SS_W + SS_{B1} + SS_{B2} + SS_{int}$
 - MANOVA 2 table: SS, degrees of freedom, distribution
 - estimators S_{B1} , S_{B2} , S_W , S_{int} for Σ
 - **SLIDE: MANOVA 2 test for systematic interaction + conclusion**
MANOVA 2 test for factor 1 (column) effect + conclusion
MANOVA 2 test for factor 2 (row) effect + conclusion
 - Model check as for MANOVA 1
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